

Social learning drives mis-prioritization of collective challenges

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Abstract

Societies often struggle to prioritize competing challenges in a timely manner, with substantial costs from delayed action on issues like climate change and pandemic mitigation. A persistent puzzle is that broad concern often fails to translate into collective priority, as with climate relative to the economy. Most models of opinion dynamics analyze issues in isolation and cannot explain how competing concerns with inherent trade-offs are reconciled into collective priority. We argue that a key driver is the differing reliance on social learning across domains. Some issues rely heavily on social learning, inferring importance from others, because direct experience is limited; others rely more on individual learning from firsthand experience, as with economic conditions. To identify endogenous mechanisms driving this mis-prioritization, we develop a dynamic model in which individuals in two subgroups form issue-specific concerns through social and individual learning, aggregated into collective priority. The model yields three insights. First, at equal objective severity, issues that depend

more on social learning are systematically mis-prioritized. Second, when social learning is high, gradual increases in severity may not raise collective priority, producing delayed action or persistent misalignment. Third, cross-group learning can mitigate this bias, but only above a critical threshold. Two levers can realign collective priority with severity. One is sustained cross-group engagement that pushes inter-group learning past the threshold. The other is strengthening firsthand experience of an issue, reducing reliance on social cues.

Significance statement

A majority of Americans report worrying about climate change, yet only a minority rank it as a top national priority. Why does broad concern so often fail to translate into collective priority? Existing models of opinion dynamics study issues one at a time and cannot answer this. We build a simple mathematical model in which people form concerns about competing issues through both personal experience and observation of others. The model shows that issues relying heavily on social learning are systematically under-prioritized even when equally severe, and that rising severity alone does not necessarily raise priority. Two interventions can correct this. Increasing inter-group contact above a tipping point, and reducing reliance on social learning by making consequences more directly observable, can together realign priority with severity.

1 Introduction

Human societies often face multiple, sometimes conflicting, challenges simultaneously. We seek economic growth while striving to mitigate climate change, or aim to curb the spread of an epidemic through social distancing while attempting to protect retail businesses. Yet efforts to balance competing priorities frequently fall short. Despite scientific consensus on the threat of anthropogenic climate change [1, 2] and a majority of Americans expressing concern [3], no consistent federal-level mitigation policy has emerged. Similarly, during the COVID-19 pandemic, partisan divisions undermined the effectiveness of containment measures such as social distanc-

ing. A recurring pattern in these failures is that collective decision-making often falters, failing to prioritize societal threats in a timely manner.

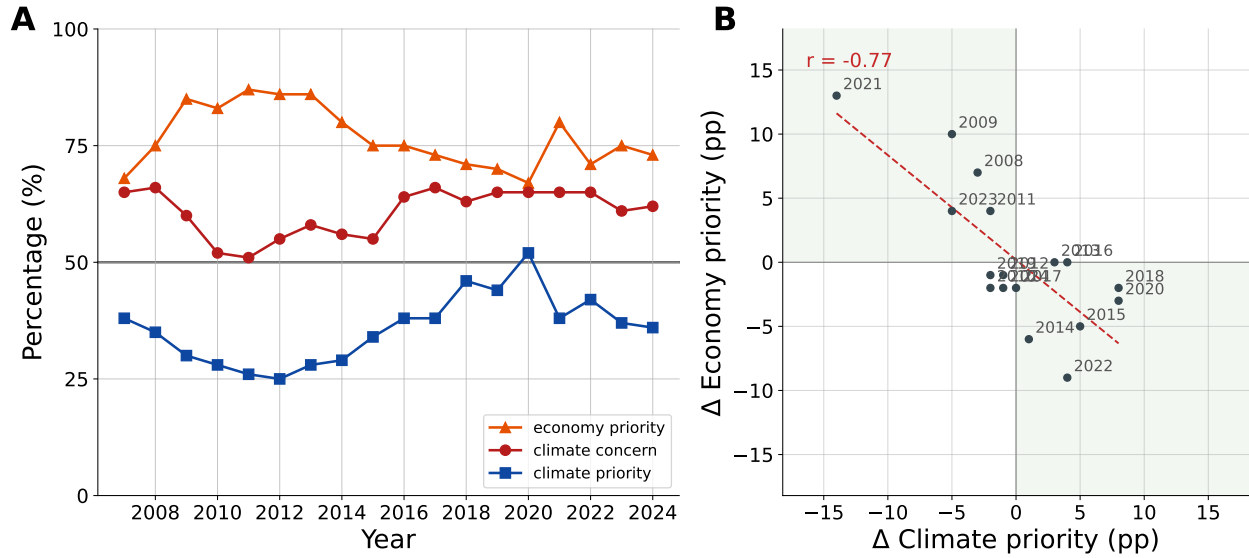


Figure 1: Majority concern does not translate into majority priority. (A) Gallup climate worry, Pew climate priority, and Pew economy priority among U.S. adults, 2007–2024. (B) Year-over-year changes in climate and economy priority ($r = -0.77$).

Figure 1 illustrates this pattern in U.S. public opinion. The survey data consistently show that roughly 60% of Americans report worrying about global warming, yet only about 35% rank climate as a top national priority, far below the economy at roughly 75% (Fig. 1A). A majority of Americans are concerned about climate change, but a minority treat it as a top governmental priority. Because climate mitigation and economic growth involve inherent trade-offs, prioritizing one necessarily comes at the cost of the other. Indeed, the data show that year-over-year changes in climate and economy priority are strongly negatively correlated ($r = -0.77$, Fig. 1B). The persistent failure to prioritize climate action has been widely attributed to exogenous forces such as partisan elite cues and strategic media framing [4, 5]. Understanding the endogenous mechanisms by which majority concern fails to translate into majority priority is the central aim of this paper.

The formation of opinions in collectives has been widely studied using dynamical models across physics, mathematics, psychology, and sociology [6–8]. Most models analyze how opin-

ions about a single issue evolve under social influence and external signals, with some extending to multi-dimensional opinion vectors [9–11]. A smaller but growing literature examines cross-issue interactions, emphasizing how associations between issues are reinforced through observing others’ behaviors [12, 13]. Yet it remains unclear how concerns for multiple issues with distinct characteristics compete for priority in a polarized society, a problem central to the emerging discourse on collective adaptation [14].

The key lies in how concerns are formed, and specifically in how the balance between social and individual learning differs across issues. Concern (or opinion) formation is typically attributed to two primary learning mechanisms, social learning and individual learning, with some models allowing a combination of both [6, 15–19]. Social learning refers to adopting the beliefs or behaviors of other people, such as by observing their behaviors or communicated concerns. Here we focus on frequency-based learning, whereby individuals disproportionately adopt the concerns expressed by the majority of their neighbors. Conformity to majority opinion is a robust and widespread phenomenon in human judgment [20], and perceived social consensus is a well-documented driver of concern formation, particularly for issues like climate change where individuals cannot directly observe outcomes [21].

By contrast, individual (or asocial) learning occurs when people interact directly with their environment and acquire information that informs their concern about an issue’s severity. Individual learning is most feasible when the environment is observable, whereas limited observability can push people toward greater reliance on social learning [22, 23]. Other empirical evidence shows that reliance on social learning increases when tasks are difficult or complex [24–27], when uncertainty is high [28–30], when confidence in one’s own judgment is low [27, 31], and when the cost of individual exploration is high [27]. Consequently, the degree of social learning should vary in concern formation across issues, reflecting issue-specific levels of complexity, uncertainty, confidence, and learning costs. We therefore model reliance on social learning as issue-specific.

Climate change and early-stage pandemics exemplify domains where social learning dominates concern formation. Climate impacts are temporally delayed and geographically distant,

limiting individual learning and elevating uncertainty [32], and acquiring scientific information about severity is costly for most individuals. Accordingly, climate concern in the United States is strongly structured by political affiliation, with partisanship and ideology frequently outweighing experiential exposure by as much as double [33, 34]. The early COVID-19 pandemic exhibited similar conditions. The virus’s invisibility, delayed feedback, and rapidly evolving science made individual learning difficult and unreliable, and concern and protective behavior were strongly shaped by ideology [35, 36]. As cases accumulated and individual learning improved, the role of social learning diminished [37].

By contrast, economic conditions are frequently observed through prices, wages, and employment, lowering uncertainty and reducing the cost of individual learning. Economic concerns are therefore shaped more by individual learning and less by social learning. Partisan bias meaningfully affects sociotropic (national) economic evaluations but has only marginal influence on assessments of household finances [38, 39], and inflation beliefs co-move strongly with household income changes and direct price experience [40, 41].

Beyond how much individuals rely on social learning, whom they learn from is equally important. In a polarized society, people tend to learn disproportionately from in-group partisans, driven in part by selective exposure to social networks dominated by co-partisans [42, 43]. Efforts to reduce polarization often aim to promote meaningful cross-party collaboration [44], encouraging consideration of out-partisan perspectives.

Here, we develop a two-group, two-issue dynamical model of collective issue prioritization that captures competition between concerns for issues with differing degrees of social learning. Our model yields three main findings. First, when two equally severe issues compete for collective attention, the issue whose concern formation relies more heavily on social learning is systematically mis-prioritized. Second, when objective severity increases gradually, high social learning delays or prevents collective re-prioritization. Third, restoring inter-group learning can correct this mis-prioritization, but only above a connectivity threshold, suggesting that modest depolarization efforts may be insufficient once concern is deeply entrenched.

2 Summary of the Mathematical Model

We consider a collective composed of two equal-sized subgroups, 1 and 2 (e.g., political parties or other salient identities), facing two competing issues, H and L . *Issue H* is abstract and distant, so individuals rely more on social learning. *Issue L* is direct and observable, so individuals rely less on social learning. The issues are in conflict in the sense that actions advancing one tend to inhibit the other. Each issue has its own public discourse that shapes perceived severity within each subgroup. Population-level concern for each issue is the average of subgroup concerns, and collective priority is determined by comparing these averages.

Each individual has a binary concern state, either concerned (1) or unconcerned (0). Let $C_{g,i}(t)$ denote the proportion of individuals in group $g \in \{1, 2\}$ who are concerned about issue $i \in \{H, L\}$ at time t . Following a standard adjustment formulation [45], concern evolves toward a target $C_{g,i}^*(t)$ with timescale τ :

$$\frac{dC_{g,i}(t)}{dt} = \frac{C_{g,i}^*(t) - C_{g,i}(t)}{\tau}. \quad (1)$$

An individual may become concerned about issue i through individual learning, with probability I_i , or through social learning, with probability $f(X_{g,i}(t); \alpha)$. The parameter s_i governs the relative weight placed on social versus individual learning for issue i . Taking the expectation over individuals yields the group-level target concern:

$$C_{g,i}^*(t) = (1 - s_i) I_i + s_i f(X_{g,i}(t); \alpha). \quad (2)$$

Here $I_i \in [0, 1]$ represents the *objective severity* of issue i , defined as the expected fraction of the population that would be concerned under purely individual learning. We assume I_i does not systematically differ across groups, since it reflects objective severity. The term $f(X_{g,i}(t); \alpha)$ represents the corresponding fraction under purely social learning, mapping the observed proportion of concerned peers $X_{g,i}(t)$ through an S-shaped conformity response function with shape parameter $\alpha > 0$ [46–48] (Figure 2B). When $s_i = 0$, concern is determined entirely by objective

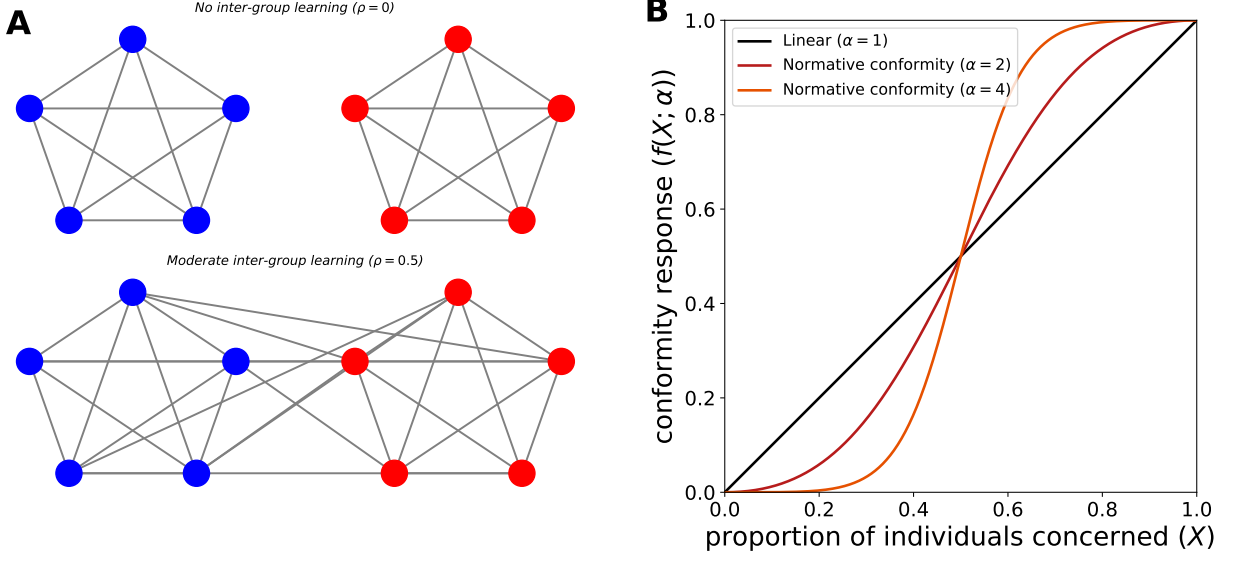


Figure 2: Model components. (A) Two-group network with full intra-group connectivity. Top panel shows no inter-group learning ($\rho = 0$) and bottom panel shows moderate inter-group learning ($\rho = 0.5$). Blue and red nodes represent the two groups. (B) Conformity response function $f(X; \alpha)$ for varying levels of normative conformity.

severity. When $s_i = 1$, it is determined entirely by the share of concerned peers.

The proportion of concerned individuals observed by a group-1 individual is

$$X_{1,i}(t) = \frac{1}{1+\rho} C_{1,i}(t) + \frac{\rho}{1+\rho} C_{2,i}(t), \quad (3)$$

where ρ is the probability of an inter-group connection in an individual's network. The expression is symmetric for group 2. As illustrated in Figure 2A, when $\rho = 0$, individuals observe only in-group concern. When $\rho = 1$, the two groups are well-mixed.

Given group-level concerns, population-level concern for issue i is $C_i(t) = \frac{1}{2}C_{1,i}(t) + \frac{1}{2}C_{2,i}(t)$. The collective prioritization of *issue H* relative to *issue L* is evaluated via a logistic choice function [45, 49–52]:

$$P(t) = \frac{\exp(C_H(t)/\sigma)}{\exp(C_H(t)/\sigma) + \exp(C_L(t)/\sigma)}, \quad (4)$$

where $\sigma > 0$ controls sensitivity to differences in concern, with smaller σ implying sharper

prioritization. This formulation reflects the assumption that aggregate priorities track whichever issue commands greater population-level concern.

3 Results

We show that social learning can systematically distort collective prioritization, examine why this distortion is difficult to correct as problems worsen, and identify conditions under which restoring inter-group communication can realign priorities with objective severity.

When two equally severe issues compete for collective attention, the issue whose concern formation relies more on social learning is mis-prioritized at equilibrium. Concern for the high-social-learning issue polarizes between groups, whereas concern for the low-social-learning issue remains anchored to objective severity through individual learning [53].

This pattern speaks to a long-standing puzzle. A majority of Americans report concern about climate change, and the scientific community agrees on its severity. Yet climate mitigation consistently loses priority to economic policy. We begin with fully disconnected groups, reflecting extreme affective polarization. Figure 3A–C show the time evolution of concern and priority for two equally severe issues, in which H relies more on social learning than L . Concern for H diverges between groups (Fig. 3C), whereas concern for L converges. A majority of the population is concerned about H (Fig. 3B), but this majority is concentrated in one group while the other remains largely unconcerned. Because concern for L is uniformly high across both groups, L attains higher collective priority despite equal objective severity (Fig. 3A). This mirrors the climate and economy case, in which broad but polarized concern fails to translate into collective priority when a competing issue commands uniform attention.

Two model parameters can shift this outcome. First, reducing social learning for issue H raises its priority to a majority (Figure 4A). Second, introducing inter-group learning dampens between-group divergence in concern for H and likewise restores a majority priority for H (Figure 4B).

Panels C and D of Figure 4 generalize these results by mapping equilibrium prioritization

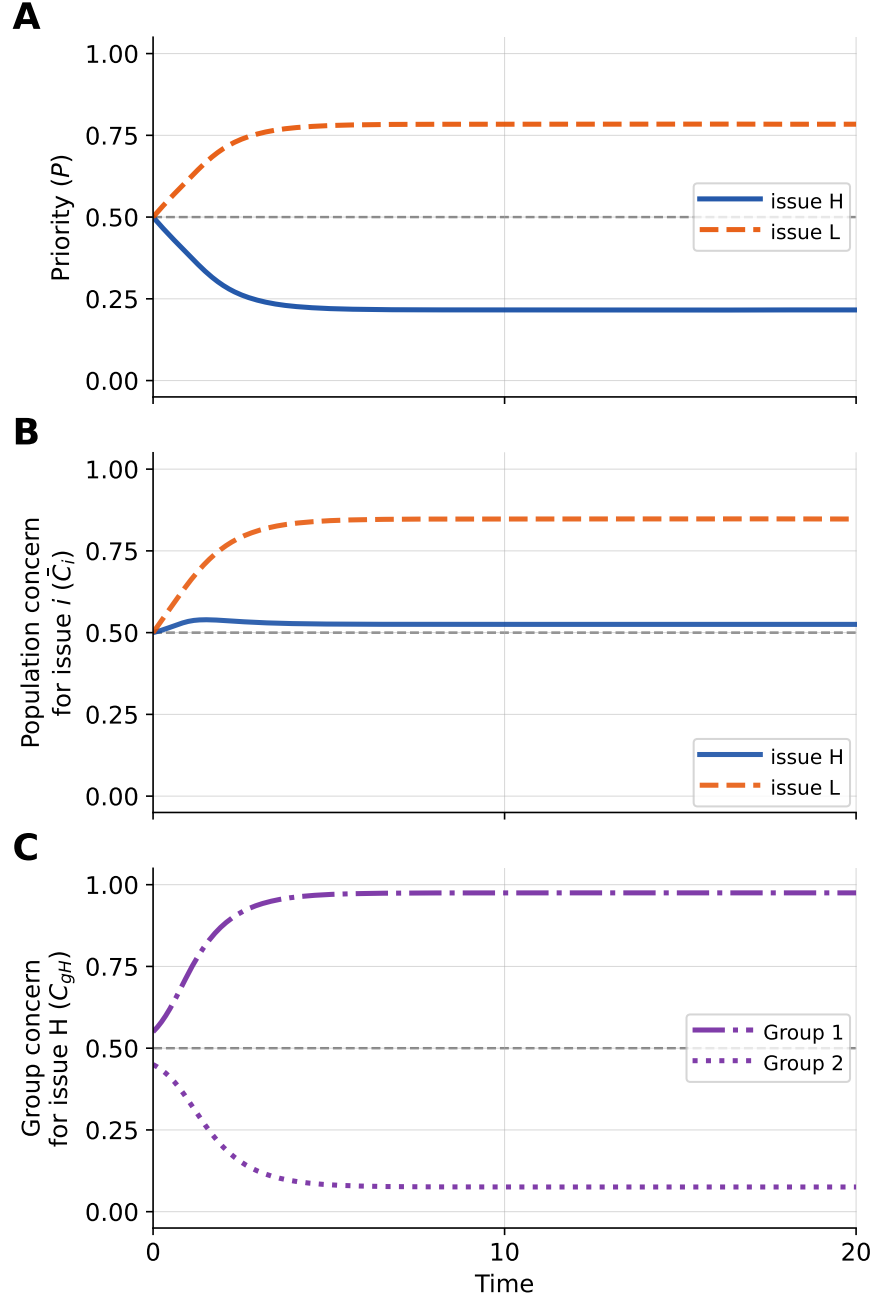


Figure 3: High social learning produces concern polarization and mis-prioritization. (A) Population priority for H and L . (B) Population-level concern for each issue. (C) Group-level concern for H , showing between-group divergence.

across social learning for H (s_H) and shared objective severity. With no inter-group learning, increasing s_H sharpens the prioritization distortion (Figure 4C). The two dots in panel C mark the baseline and reduced- s_H equilibria from panel A, showing that the minority-to-majority transi-

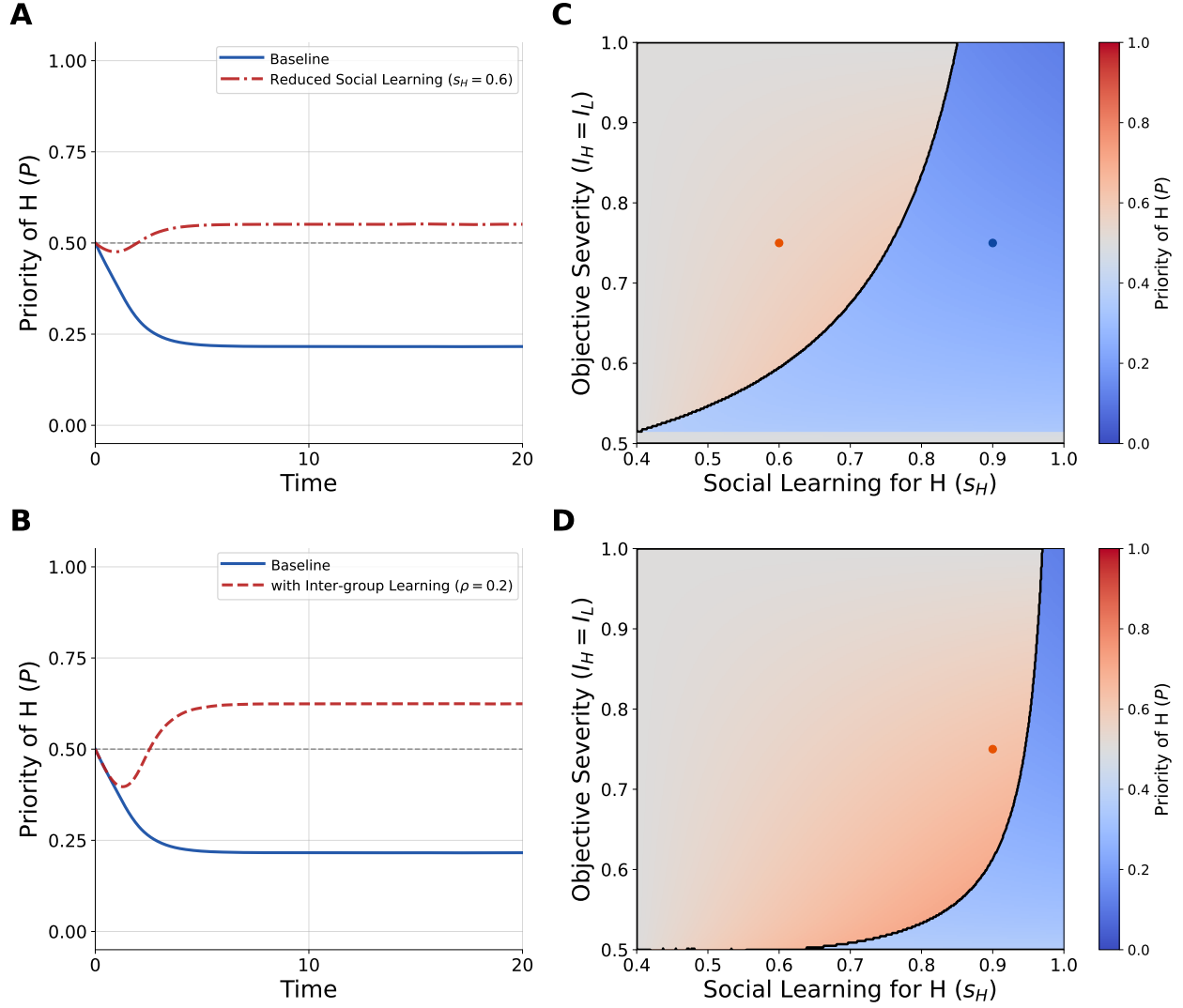


Figure 4: Both inter-group learning and reduced social learning can restore prioritization of H . (A) Reducing social learning for H raises its priority. (B) Introducing inter-group learning raises priority for H . (C, D) Equilibrium priority across social learning for H (s_H) and shared objective severity, without inter-group learning (C) and with inter-group learning (D). Dots mark equilibria corresponding to A and B.

tion corresponds to crossing the boundary along the s_H axis at fixed severity. Above a critical s_H , H is under-prioritized regardless of objective severity. Introducing inter-group learning shrinks the mis-prioritization region (Figure 4D), so the parameter point that lies in the minority region of panel C falls in the majority region of panel D. Inter-group learning therefore improves collective prioritization across a wider range of conditions.

The objective severity axis in panels C and D is restricted to the upper half, capturing issues

for which a majority of the population would be concerned under purely individual learning. This regime is most relevant to competition among severe societal challenges. At lower severity the pattern reverses, and H is over-prioritized through social reinforcement. The full heatmap covering the complete severity range appears in the Supplementary Information (Figure S1).

When objective severity rises gradually, high social learning delays or prevents collective re-prioritization. Strong social learning anchors the population in a low-concern equilibrium early, and this state persists as severity grows. Figure 5 shows objective severity for H increasing over time while L remains fixed. At moderate social learning, priority tracks severity with a modest lag. At higher social learning, this lag grows substantially. At extreme social learning, priority fails to reach a majority even after severity reaches its maximum.

This outcome is path-dependent. When severity is high from the outset, the population sustains appropriate prioritization. When the same severity level is reached gradually, early low-concern equilibria persist through social reinforcement, leaving the population under-prioritizing the issue. Gradually worsening issues that rely heavily on social learning are therefore especially vulnerable to sustained mis-prioritization.

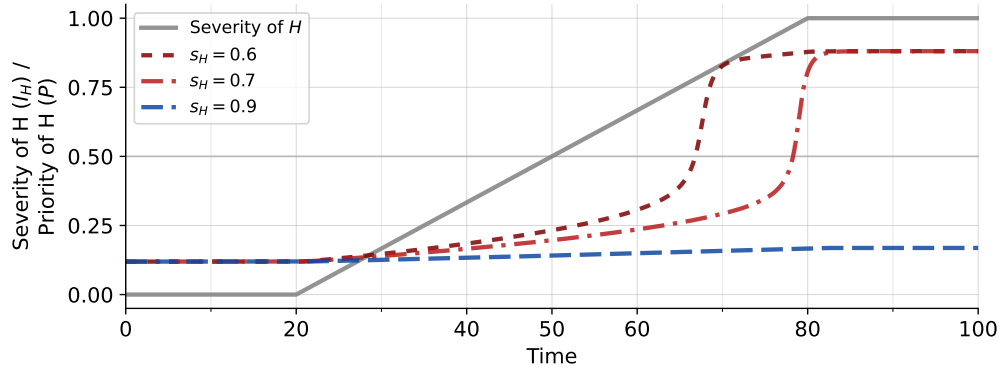


Figure 5: Higher social learning delays or prevents prioritization of a rising-severity issue. The solid line shows objective severity for H rising linearly over time, with severity for L held constant. Dashed lines show the resulting priority of H over L at three levels of social learning.

Interventions that bridge partisan divides, such as cross-party dialogue initiatives, aim to restore communication between ideologically opposed groups. We examine how much cross-group exposure is required to correct mis-prioritization once concern is already polarized. Figure 6 com-

197 pares two scenarios in which inter-group learning increases to either a small or a large value at
 198 $t = 50$. Before the change, both scenarios share the same equilibrium: concern for H is polar-
 199 ized, concern for L is uniformly high, and H is under-prioritized. A small increase in inter-group
 200 learning does not reduce this polarization. A large increase causes both groups to converge on
 201 high concern for H , shifting priority to a majority (Supplementary Information, Fig. S2 shows the
 202 underlying group-level concern dynamics for both issues). Inter-group connectivity must there-
 203 fore exceed a threshold to realign prioritization with objective severity, implying that modest
 204 depolarization efforts may be insufficient once concern is entrenched.

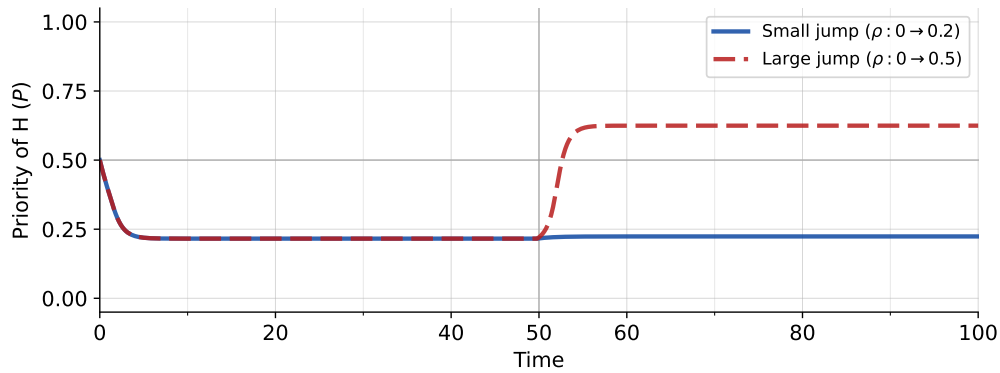


Figure 6: Restoring inter-group connectivity corrects mis-prioritization only above a threshold. The two scenarios differ in the size of the increase in inter-group learning at $t = 50$.

205 4 Discussion

206 Our results show how issue characteristics and social learning jointly shape collective prioritiza-
 207 tion between competing issues. When issues are equally severe, social dynamics lead groups
 208 to systematically under-prioritize the one that relies more heavily on social learning. When
 209 both issues are equally mild, the pattern reverses as individuals rely more on social cues and
 210 in-group conformity. When objective severity increases gradually, collective prioritization of
 211 a high-social-learning issue can shift only after a substantial delay or fail to adjust altogether,
 212 even when severity becomes extreme. When inter-group connectivity increases from a polarized
 213 baseline, it must exceed a threshold before concerns converge and priorities realign with severity.

Below that threshold, cross-group interaction produces only modest reductions in polarization. Notably, this bias is distinct from individual-level cognitive biases. Our model assumes that each individual, when using individual learning, accurately perceives objective severity, yet collective mis-prioritization still emerges from social learning dynamics alone.

The model’s predictions align with the puzzling patterns in U.S. public opinion. Counterintuitively, climate change mitigation has been de-prioritized behind economic policies in spite of the scientific consensus of the severity of anthropogenic climate change and majority concern in the US. The gap between majority concern for climate change and its minority standing as a national priority, relative to the economy (Figure 1A), is what the model produces when a high-social-learning issue competes with a low-social-learning issue of comparable objective severity. The strong negative correlation between year-over-year changes in climate and economy priority ($r = -0.77$, Figure 1B) further supports the competitive trade-off that the model’s priority function captures. The present model is built on inherent policy trade-offs, where actions taken to mitigate one issue directly exacerbate the other, as when emissions reduction raises energy costs and slows near-term economic growth. However, the underlying logic may generalize to issues that are not inherently in conflict, because individuals also face a finite cognitive budget for attention [54, 55]. To the extent that attention allocated to one issue reduces attention available for others, similar mis-prioritization dynamics could emerge even between issues whose objectives do not directly oppose each other.

The model also predicts that collective responses to different issues can unfold at markedly different speeds, and in some cases fail to occur at all. When objective severity increases gradually for a high-social-learning issue, concern and priority may adjust only after a substantial delay or remain misaligned indefinitely, depending on the strength of social learning and inter-group connectivity (Figure 5). The model suggests that such variation can arise in part from differences in how much concern formation depends on social learning relative to direct experience. Importantly, the responsiveness of collective adjustment may also vary across communities, reflecting differences in social network structure and cultural tightness [56] that shape how strongly indi-

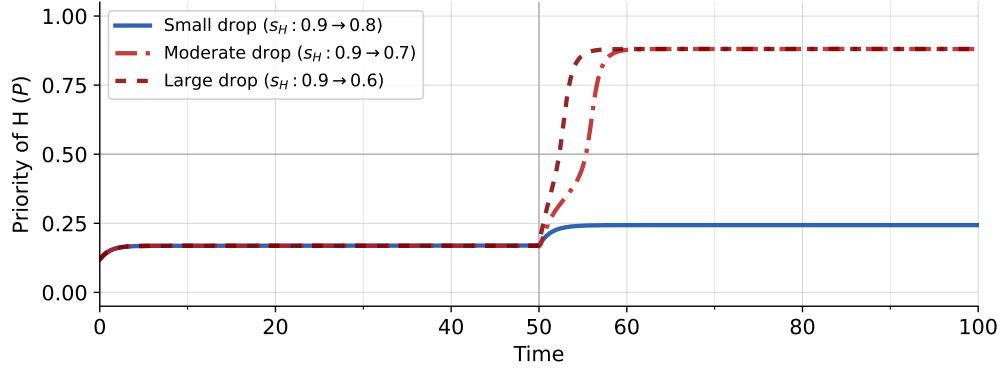


Figure 7: Reducing social learning can shift collective priority toward the high-social-learning issue. Starting from the high-social-learning equilibrium in Figure 5, s_H is reduced at $t = 50$. Larger reductions produce more responsive shifts in priority.

viduals rely on in-group cues.

A further implication concerns politicization. When a high-social-learning issue becomes tied to group identity and inter-group learning is near zero, within-party advocacy alone cannot shift collective prioritization. If the science-communication environment turns climate beliefs into markers of cultural identity [57], advocacy within one party can reinforce rather than reduce the concern gap. Restoring inter-group learning is a prerequisite for collective reprioritization, unless both parties independently arrive at concern through separate channels. This contrasts with the ozone-depletion episode, which was resolved amid lower politicization, greater immediate salience, aligned elite messaging, and the availability of commercially viable chlorofluorocarbon (CFC) substitutes [58].

The model highlights two levers for shifting collective priorities. The first is increasing inter-group connectivity beyond the threshold identified above. When baseline connectivity is very low, small increases have little effect, but sustained increases can eventually cross a tipping point, producing a sudden and large shift in concern and collective priorities. This has implications for cross-partisan engagement programs [44, 59], which may show limited measurable impact initially but could trigger rapid change once a critical level of inter-group interaction is reached.

The second lever is reducing the degree to which concern formation depends on social learning. Figure 5 showed that when social learning is high, collective priority can remain suppressed

even as objective severity rises. Starting from that outcome, Figure 7 shows that reducing s_H can unlock collective reprioritization, with larger reductions producing faster and more complete shifts. This suggests that interventions making the consequences of a threat more immediate and tangible, such as interactive climate simulations [60], can complement cross-partisan engagement by shifting the basis of concern from social learning toward individual learning. These two levers operate through distinct channels, and combining both may be more effective than either alone.

The model is intentionally parsimonious to isolate the effects of the key social dynamics. It does not model awareness diffusion, which future work could integrate with the concern-formation dynamics studied here. The framework best fits contexts in which two issues are inherently in conflict in the near term, such as climate mitigation versus economic growth. Over longer horizons these trade-offs can become synergistic, as when the Inflation Reduction Act of 2022 bundled climate investment with job creation and energy cost reduction. We model two groups to capture two-party concern polarization in the United States, though multi-party systems in which coalition formation yields a governing bloc and an opposition can be mapped into the same structure. We model two issues because clear direct trade-offs among three or more issues are rarer in practice. Finally, the model assumes frequency-based social learning with homogeneous influence within groups. Incorporating expert learning, heterogeneous credibility, and real-world network structure is an important direction for future work.

Our work makes two contributions. First, we offer an endogenous explanation for patterns widely observed in polarized societies, such as the persistent gap between climate concern and climate prioritization, without attributing these outcomes to media distortion or elite manipulation. The mis-prioritization instead emerges from the structure of social learning itself, operating through in-group and out-group dynamics. Second, we extend the opinion dynamics literature from single-issue belief formation to multi-issue priority-setting under competing demands. Most formal models analyze how opinions about a single issue evolve under social influence [52, 61], but real-world challenges rarely arise in isolation and often compete. Recent work has called

for formal models that address how societies manage multiple competing challenges, notably the collective adaptation framework [14]. Our model responds to this call by making it possible to study when societies adapt to changing severity across domains and when they mis-prioritize despite rising objective severity.

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5 Supplementary Information

5.1 Data sources

Figure 1 draws on two U.S. public opinion survey series. Climate concern data are from the Gallup Poll Social Series: Environment, an annual March survey of approximately 1,000 U.S. adults. We use the combined share reporting that they worry “a great deal” or “a fair amount” about global warming or climate change. Policy priority data are from Pew Research Center, which asks whether each of a set of issues should be a “top priority” for the President and

Congress. We use the share rating “dealing with global climate change” and “strengthening the nation’s economy” as top priorities. The series was administered by phone through 2020 ($N \approx 1,500$) and via the American Trends Panel online from 2021 onward ($N \geq 2,500$ per item). The mode change may introduce a level shift. Toplines are available from Gallup (<https://news.gallup.com/poll/1615/environment.aspx>) and Pew Research Center (<https://www.pewresearch.org/topic/politics-policy/>).

5.2 Parameter values and rationale

Table S1 lists the baseline parameter values, and Table S2 records the parameters used in each figure. These values illustrate the model’s qualitative regimes rather than fit any specific empirical case. The equilibrium maps in Figure 4C–D and Figure S1 show that these regimes persist across a broad swath of parameter space.

The social learning rates $s_H = 0.9$ and $s_L = 0.4$ instantiate the model’s central contrast between an issue that relies more on social learning and one that relies more on individual learning. Initial group concerns are small symmetric perturbations around 0.5, allowing within-group reinforcement to amplify slight asymmetries. The direction of symmetry is reversed across the two groups for the two issues, reflecting groups with slightly different initial concerns. The conformity exponent $\alpha = 3$ yields a moderately S-shaped response consistent with empirical estimates of frequency-based conformity [46, 48], and the choice sensitivity $\sigma = 0.25$ produces prioritization responsive to concern differences without being deterministic. Objective severities $I_H = I_L = 0.75$ place both issues in the high-severity regime relevant to the climate–economy case (Figure 1A). Inter-group connectivity $\rho = 0$ represents a maximally polarized baseline, and $t_{\text{final}} = 20$ is well past equilibrium for baseline simulations.

Table S1: Baseline parameter values used in Figure 3. All other figures inherit these values except where Table S2 specifies otherwise.

Parameter	Value	Description
C_0	(0.55, 0.45, 0.45, 0.55)	Initial group concerns, $(C_{1H}, C_{2H}, C_{1L}, C_{2L})$
s_H, s_L	0.9, 0.4	Social learning weight (issues H, L)
I_H, I_L	0.75, 0.75	Objective severity (issues H, L)
ρ	0	Inter-group connectivity
σ	0.25	Logit choice sensitivity
α	3	Conformity exponent
t_{final}	20	Integration horizon

Table S2: Parameter values for each figure that differ from the baseline in Table S1. Parameters not listed are at baseline.

Figure	Changes from baseline
Fig. 3	baseline (no changes)
Fig. 4A	overlay: $s_H = 0.6$ vs. 0.9
Fig. 4B	overlay: $\rho = 0.2$ vs. 0
Fig. 4C	$s_H \in [s_L, 1], I_H = I_L \in [0.5, 1]$ swept; equilibria computed by fixed-point iteration
Fig. 4D	as Fig. 4C with $\rho = 0.2$
Fig. 5	$C_0 = (0, 0, 0.5, 0.5), s_H \in \{0.6, 0.7, 0.9\}, I_H(t) = \min(\max((t - 20)/60, 0), 1), I_L = 0.5, t_{\text{final}} = 100$
Fig. 6	$\rho(t) = 0$ for $t < 50; \rho \in \{0.2, 0.5\}$ for $t \geq 50; t_{\text{final}} = 100$
Fig. 7	$C_0 = (0, 0, 0.5, 0.5), s_H(t) = 0.9$ for $t < 50$ and $s_H \in \{0.8, 0.7, 0.6\}$ for $t \geq 50; I_H = 1.0, I_L = 0.5, t_{\text{final}} = 100$
Fig. S1A	as Fig. 4C, swept $I \in [0, 1]$
Fig. S1B	as Fig. 4D, swept $I \in [0, 1]$
Fig. S2	as Fig. 6

5.3 Robustness

[TBD]

5.4 Additional results

Figure S1 extends the equilibrium maps of Figure 4C–D to the full severity range $I_H = I_L \in [0, 1]$. The upper halves of the panels reproduce the main-text maps, where H is under-prioritized at high social learning. At low severity ($I_H = I_L < 0.5$), the pattern reverses, and H is over-prioritized through social reinforcement when only a minority would be concerned under purely individual learning. Inter-group learning shrinks the mis-prioritization region in both directions.

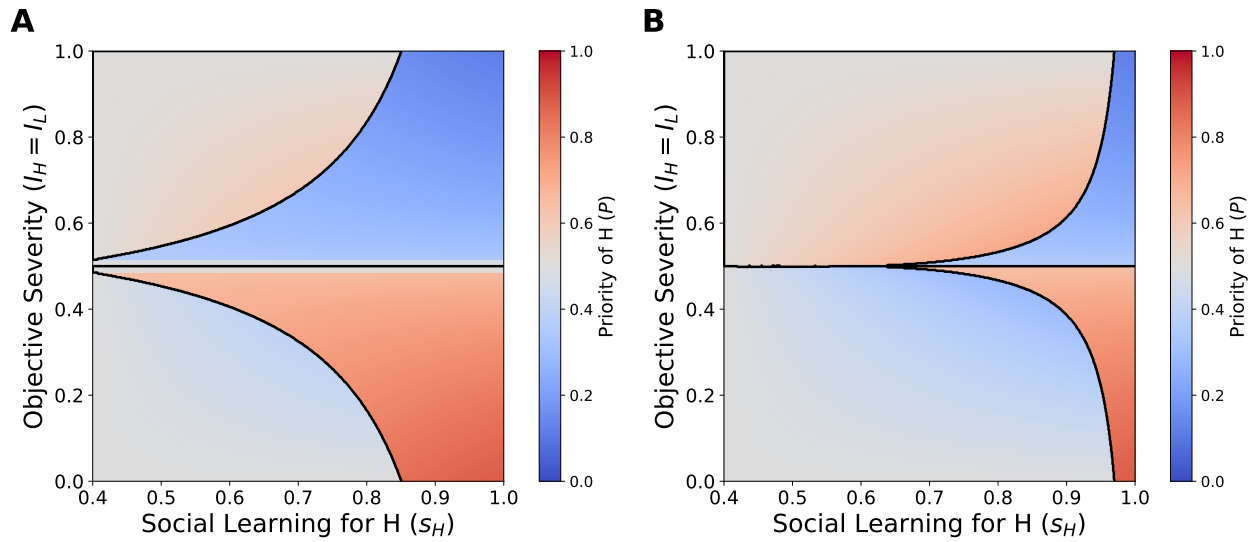


Figure S1: Equilibrium priority of H across the full range of objective severity, extending Figure 4C–D. (A) Without inter-group learning ($\rho = 0$). (B) With inter-group learning ($\rho = 0.2$). Upper halves of the panels appear in the main text. Lower halves show that at low severity ($I_H = I_L < 0.5$), H is over-prioritized through social reinforcement, the mirror image of the high-severity regime.

Figure S2 displays the group-level concern dynamics underlying the threshold behavior in Figure 6. Before the perturbation at $t = 50$, concern for H is polarized between groups while concern for L is uniformly high. The small jump in inter-group learning leaves the polarization in concern for H largely intact, while the large jump collapses it, producing the priority shift seen in panel A. Concern for L is unaffected by either intervention.

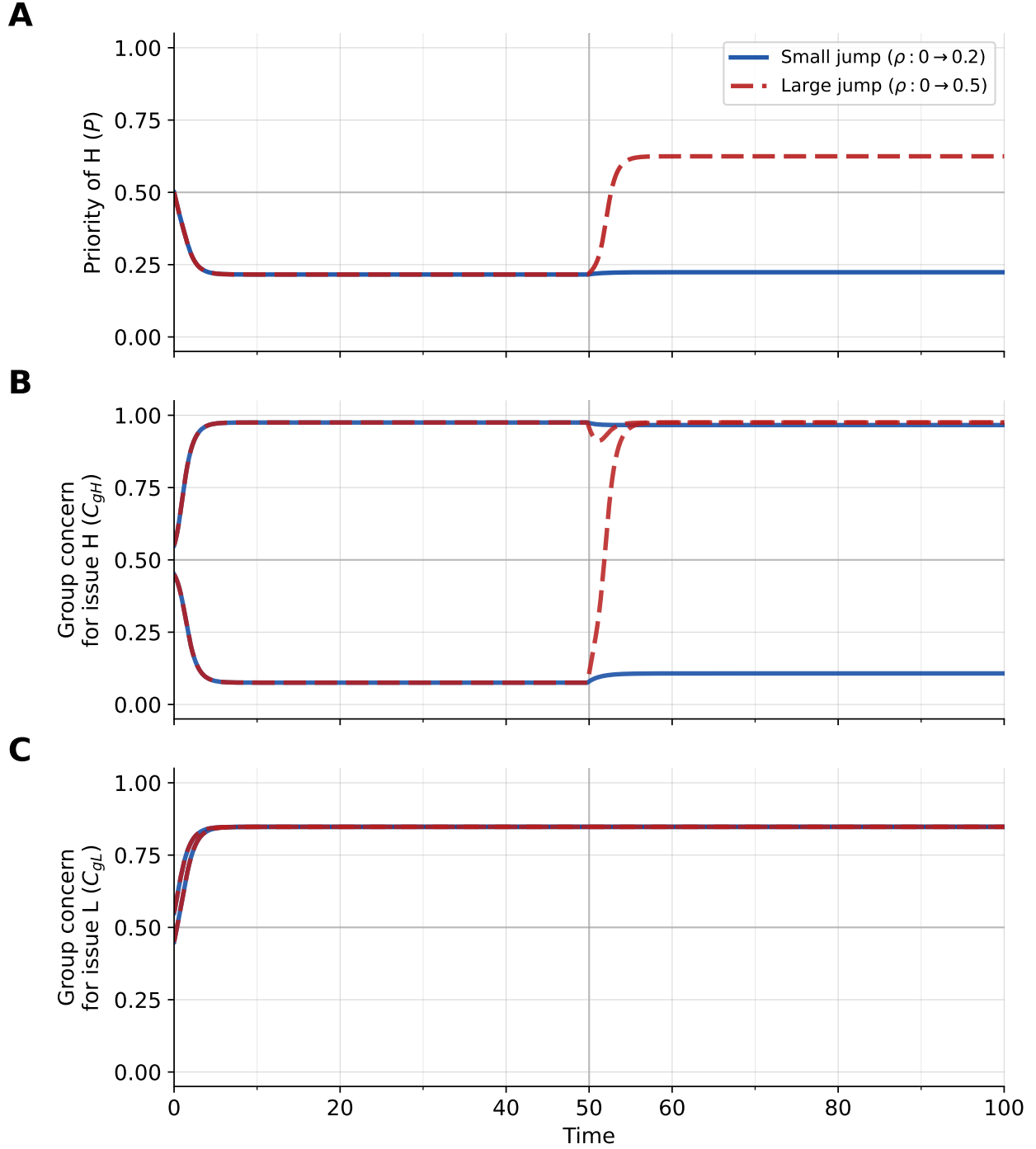


Figure S2: Group-level dynamics underlying Figure 6. At $t = 50$, inter-group connectivity increases from $\rho = 0$ to either 0.2 (small jump) or 0.5 (large jump). (A) Population priority for H over L . (B) Group-level concern for issue H in each group, showing that only the large jump collapses between-group divergence. (C) Group-level concern for issue L , which remains uniformly high in both scenarios.